

2025

COMPUTER SCIENCE AND ENGINEERING

Paper : CSOE-1102

(Operation Research)

Full Marks : 70

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer *question nos. 1 & 2* and *any four* questions from the rest.

1. Answer *any five* questions :

2×5

- (a) Define a Linear Programming Problem and mention its two basic components.
- (b) What is a balanced transportation problem?
- (c) State the main difference between a transportation problem and an assignment problem.
- (d) What is a two-person zero-sum game?
- (e) What is meant by the saddle point in a game?
- (f) What is the purpose of Vogel's Approximation Method (VAM)?
- (g) Differentiate between PERT and CPM (Mention any two points).

2. Answer *any five* questions :

4×5

- (a) Food X contains 6 units of vitamin A/gr and 7 units of vitamin B/gr and costs 12 paisa/gr. Food Y contains 8 units of vitamin A/gr and 12 units of vitamin B/gr and costs 20 paisa/gr. The daily vitamin requirement of vitamin A and vitamin B is 100 units and 120 units, respectively. Find the minimum cost of the product mix by formulating the problem.

- (b) Construct the dual to the primal problem :

$$\text{Max } Z = 3x_1 + 5x_2$$

$$\text{s/t, } 2x_1 + 6x_2 \leq 50$$

$$3x_1 + 2x_2 \leq 35$$

$$5x_1 - 3x_2 \leq 10$$

$$x_2 \leq 20,$$

$$\text{where } x_1, x_2 \geq 0.$$

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(c) Solve the Transportation problem using North-West corner rule :

Source/Dest.	1	2	3	4	Supply
1	10	0	20	11	15
2	12	7	9	20	25
3	0	14	16	18	5
Demand	5	15	15	10	

(d) Solve the two person zero sum game :

Player A		
Player B	2	-1
	-1	1

(e) Solve the following assignment problem :

	J ₁	J ₂	J ₃
A	26	23	27
B	23	22	24
C	24	20	23

(f) Find the critical path of the following characteristics :

Activity	1-2	1-3	2-4	3-4	3-5	4-9	5-6	5-7	6-8	7-8	8-10	9-10
Time (days)	4	1	1	1	6	5	4	8	1	2	5	7

(g) Discuss Optimistic time estimation, Most likely time estimation, Pessimistic time estimation and their mathematical relationship.

3. Solve the following LPP by the Simplex Method :

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$$\text{Max } Z = 2x_1 + 5x_2$$

$$\text{s/t, } x_1 + 4x_2 \leq 24$$

$$3x_1 + x_2 \leq 21$$

$$x_1 + x_2 \leq 9$$

$$x_1, x_2 \geq 0.$$

4. Solve the following Transportation problem with the optimality test :

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						Source
	7	6	4	5	9	40
	8	5	6	7	8	30
	6	8	9	6	5	20
	5	7	7	8	6	10
Destination	30	30	15	20	5	

5. The following table shows the jobs of a network along with their time estimate :

Job	1-2	1-6	2-3	2-4	3-5	4-5	6-7	5-8	7-8
a (days)	1	2	2	2	7	5	5	3	8
m (days)	7	5	14	5	10	5	8	3	17
b (days)	13	14	26	8	19	17	29	9	32

Draw the project network and find the probability that the project is completed in 40 days.

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6. What do you mean by two person zero sum game? Write the formula of mixed strategy. Using the principle of dominance, solve the following game : 2+2+6

Player A	Player B		
	3	-2	4
	-1	4	2
	2	2	6

7. Solve the following LPP by Big M Method :

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$$\text{Min } Z = 12x_1 + 20x_2$$

$$\text{s/t, } 6x_1 + 8x_2 \geq 100$$

$$7x_1 + 12x_2 \geq 120$$

$$x_1, x_2 \geq 0.$$

8. Use Branch and Bound technique to solve the following :

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$$\text{Max } Z = x_1 + 4x_2$$

$$\text{s/t, } 2x_1 + 4x_2 \leq 7$$

$$5x_1 + 3x_2 \leq 15$$

$$x_1, x_2 \geq 0 \text{ and are integers. (Solve it up to two sub-problems)}$$